

Midterm Study Guide

CS 472/572: Unsupervised Machine Learning (Spring'26)

Use this guide to identify key concepts and focus areas for each topic.

Clustering

A) Core idea & workflow

- Understand what objective K-means minimizes and how iterative assignment and centroid updates drive the algorithm toward a solution.
- Be familiar with convergence behavior and the computational cost associated with each run.

B) Initialization & K-means++

- Know the risks of poor centroid initialization and why it can lead to suboptimal results.
- Understand the motivation behind K-means++ and how its seeding strategy improves upon random initialization.

C) Choosing K

- Understand how the elbow method works and what its limitations are when the plot does not show a clear inflection.
- Know alternative justifications for selecting a value of K when the elbow is ambiguous.

D) Preprocessing & data issues

- Understand why feature scaling matters for distance-based methods and how unscaled features can distort results.
- Know the concept of the curse of dimensionality and its implications for distance-based clustering in high-dimensional spaces.

E) Strengths & limitations

- Be able to articulate the practical strengths of K-means alongside its key weaknesses for a non-technical audience.

- Understand why certain data shapes and the presence of outliers make K-means inappropriate.

F) Variants for other data types

- Know when K-means is conceptually unsuitable and what alternative approach is used for categorical data.
- Understand how K-modes differ from K-means in terms of the cluster representative and the distance measure used.

G) Robustness & medoids

- Understand how K-medoids differ from K-means and why it offers greater robustness.
- Know the trade-off involved and the situations where medoids are preferred over means.

H) Practical details

- Knowing the formal properties a distance measure should satisfy and why they matter in practice.
- Understand the per-iteration computational cost of the assignment and update steps.

I) Applications & intuition checks

- Understand how varying K qualitatively changes the output in an application such as image segmentation.
- Know practical safeguards (e.g., multiple restarts, smart initialization, preprocessing) that improve solution quality without sacrificing speed.

J) Comparing outcomes & explaining choices

- Be able to qualitatively compare two clustering produced by different initializations using criteria such as compactness, interpretability, and stability.
- Be able to explain your full clustering pipeline—distance choice, scaling, K selection, and variant used—to a non-expert audience.

GMM & EM

A) What a GMM is

- Know the three sets of parameters that define a Gaussian Mixture Model and the role each plays in describing the data distribution.
- Understand the latent variable interpretation of a GMM and how it relates to familiar ideas from K-means.

B) Responsibilities (soft assignments)

- Understand what a responsibility represents and why soft assignments are more informative than hard cluster labels.
- Know the structure of the responsibility formula and be able to interpret the contribution of the prior and the likelihood.
- Understand how changes to a component's mixing weight affect its responsibilities relative to other components.

C) EM algorithm intuition

- Understand the purpose of each step in EM (E-step and M-step) and why they are alternated rather than solved jointly.
- Know what it means for EM to be a coordinate-ascent procedure and what quantity it is optimizing.
- Be familiar with sensible initialization strategies and understand why initialization affects the final solution.
- Know how component parameters can be estimated from soft assignments without hard labels.

D) M-step updates

- Understand the concept of effective membership and how it drives the parameter update equations.
- Know why the updated component mean is a responsibility-weighted average and what that weighting implies.
- Understand why covariance estimation can become numerically unstable and why regularization is needed.

E) Overfitting & regularization

- Know the variance collapse failure mode and understand why the likelihood can become unbounded.
- Be familiar with at least two practical strategies to prevent this failure mode and the intuition behind each.
- Understand why overfitting in high-dimensional settings (e.g., text) is especially difficult to detect.

F) Relation to K-means

- Know the assumptions under which EM soft assignments reduce to the hard assignments used by K-means.
- Understand the limiting relationship between GMM/EM and K-means as component variances shrink toward zero.

G) Model selection & convergence behavior

- Know what to monitor during EM training and understand what kind of optimum EM is guaranteed to reach.
- Be able to compare two EM solutions obtained from different initializations and choose between them.
- Understand why initialization quality has an outsized effect on both convergence speed and final model quality.

Hierarchical Clustering & Linkages

A) Core ideas

- Know what a hierarchical clustering algorithm produces as output and how a flat partition is extracted from it.
- Understand the difference between agglomerative (bottom-up) and divisive (top-down) approaches.
- Understand why the choice of inter-cluster distance is the central design decision in agglomerative clustering.

B) Dendrogram

- Know how to read a dendrogram: what merge heights represent and what you obtain by cutting at a given height.
- Understand how small changes in the cut threshold can change the number of resulting clusters.

C) Agglomerative algorithm

- Be able to walk through the high-level steps of the agglomerative algorithm and identify what is updated after each merge.
- Understand why applying different linkage criteria to the same dataset can produce very different trees.

D) Linkage behaviors

- Know the chaining tendency of single linkage: what it looks like, why it happens, and when it might be acceptable.
- Understand the compactness bias of complete linkage and the trade-off it makes relative to single linkage.
- Understand what “balanced” means for average linkage and how it affects cluster shapes.
- Know how Ward’s method uses variance to guide merges and what cluster geometry it tends to produce.

E) Reading & using worked examples

- Be able to follow a step-by-step agglomerative example, understanding why each merge is chosen and how the distance matrix evolves.
- Know how to reason about the number of clusters produced by a given cut threshold under a specific linkage rule.

F) Picking a method & distance

- Be able to match data structure characteristics (e.g., elongated shapes, compact blobs) to the most appropriate linkage method.
- Understand why the choice of point-to-point distance metric must be considered before selecting a linkage strategy.

G) From tree to decision

- Know how to select a defensible cut height from a dendrogram, using the structure of the tree (e.g., large gaps between mergers) to justify your choice.
- Be able to qualitatively compare two clustering solutions from different linkages and reason about which is more appropriate.

H) Practicalities & tooling

- Know when Ward linkage is appropriate in scikit-learn and what constraint it places on the distance metric.
- Understand the additional insight a full dendrogram (e.g., from SciPy) provides beyond a flat label assignment.

Principal Component Analysis (PCA)

A) Motivation & purpose

- Understand what PCA is designed to achieve and why dimensionality reduction is useful in machine learning pipelines.
- Know the key reasons PCA is applied: reducing computational cost, removing redundancy, enabling visualization, addressing overfitting, and compressing data.
- Understand the general conditions under which PCA is especially effective, particularly when features are highly correlated.

B) Key terminology & mathematical foundations

- Know the definitions of variance and covariance and understand what each measure is about a dataset.
- Understand the structure and meaning of a covariance matrix: what the diagonal entries represent versus the off-diagonal entries, and why it is always symmetric.
- Know what eigenvectors and eigenvalues are conceptually and understand the relationship between them and a matrix transformation.
- Understand eigen-decomposition at a high level: what it does to a matrix and why it is central to PCA.

D) Choosing the number of components (k)

- Know the three main strategies for selecting k: variance threshold (e.g., retain 95% of variance), the scree plot elbow method, and fixing k to a target dimensionality.
- Understand the concept of explained variance ratio and how it is computed from eigenvalues.
- Know how a scree plot is interpreted and what the "elbow" indicates about the relative importance of components.

F) Geometric intuition

- Understand PCA geometrically: centering the data, rotating the coordinate system to align with directions of maximum variance, and projecting onto those axes.
- Know why the first principal component captures the most variance and each subsequent component capture the most remaining variance while remaining orthogonal to all previous ones.
- Understand why the variables in the reduced PCA representation are uncorrelated by construction.

G) Multivariate Gaussian & covariance connection

- Understand the relationship between the covariance matrix and the shape of a multivariate Gaussian distribution.
- Know how eigen-decomposition of a covariance matrix reveals the principal axes and spreads of that distribution.

Simulation by hand:

1. Agglomerative Algo (Single and Complete Linkage)
2. K-means/medoids
3. PCA Analysis